

"VSEPR Theory"

• Postulates of VSEPR Theory:-

⇒ Given by Sidgwick and Powell
in 1940.

Main assumptions:- Shape of covalent molecules is based on number of electron pairs around nucleus of central polyvalent atom (with more than one electron in valence shell) in a covalent molecule.

⇒ Nyholm and Gillespie:- Formulated this idea as theory.

1. No. of bond pairs (being shared b/w bonded atoms) and lone pairs (un-shared electron of central atom) determine geometry of molecules.

Bond pair :- (Bonding electron pair).

Lone pair :- Non-Bonding electron pair.

2. Electron pairs arrange themselves around central atom as far as

possible to avoid their repulsion.

3. Lone pair occupies more space than bond pair, because it is under the influence of only one nucleus.

Due to application on both ends, cloud of bond pair is stretched.

4. Electron pairs have order of repulsion (vander waal's repulsion), as,

Bond pair - Bond pair < Bond pair - Lone pair < Lone pair - Lone pair.

5. Two electron pairs of double bond has more electron density than one electron pair of a single bond. Three electron pairs of triple bond has more electron density than double bond. But they have same affect in determining the geometry of molecule as a single bond.

bond and three electron pairs of triple bond as considered as one electron pair of single bond.

Types:-

Type of molecules depending on No. of electron pairs.

Let:-

Central atom = A

Bonded atom = B

Lone pairs = E

We have molecules:-

Type I:-

- (a) Molecules containing two electron Pairs = AB_2

- (b) Molecules containing one lone Pair = ABE

Type II:-

- (a) Molecules having three electron pairs = AB_3

- (b) Molecules having one lone pair = AB_2E

Type III:-

(a) Molecules containing four electron pairs = AB_4

(b) Molecules containing one lone pair = AB_3E

(c) Molecules containing two lone pairs = AB_2E_2

Type IV:-

Molecules containing five electron pairs = AB_5

(a) Molecules containing no lone pair = AB_5

(b) Molecules containing one lone pair = AB_4E

(c) Molecules containing two lone pairs = AB_3E_2

(d) Molecules containing three lone pairs = AB_2E_3

Type V:-

Molecules containing six electron pairs = AB_6

(a) Molecules containing no lone pair

(b) Molecules containing one lone pair = AB_5E

(c) Molecules containing two lone pairs = AB_4E_2

(d) Molecules containing three lone pairs = AB_3E_3

Type VI:-

Molecules with seven electron pairs = AB_7

(a) Molecules with no lone pair = AB_7

(b) Molecules with one lone pair = AB_6E

(c) Molecules with two lone pairs = AB_5E_2

(d) Molecules with three lone pairs = AB_4E_3

Type I:-

(a) Molecules containing two electron pairs with no lone pair:-

Symbol:- AB_2

No. of electron pair = 2

No. of bond pair = 2

No. of lone pair = 0

Lewis Structure:- $B \times \cdot A \cdot \times B$

VSEPR Structure:- $B - A - B$

Bond angle:- 180°

Geometry:- Linear

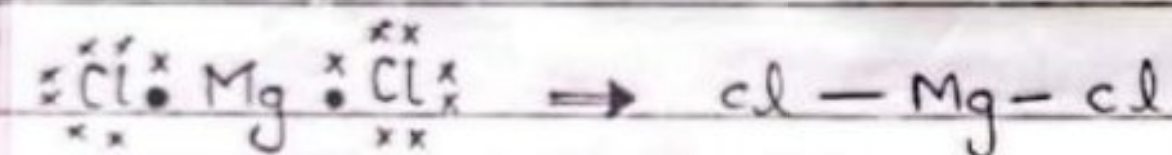
Example:- Group IIA elements.

$BeH_2, MgCl_2, CdCl_2, HgCl_2$

(i) $MgCl_2$:-

E.C of $Mg_{(12)}$:- $1s^2, 2s^2, 2p^6, 3s^2$

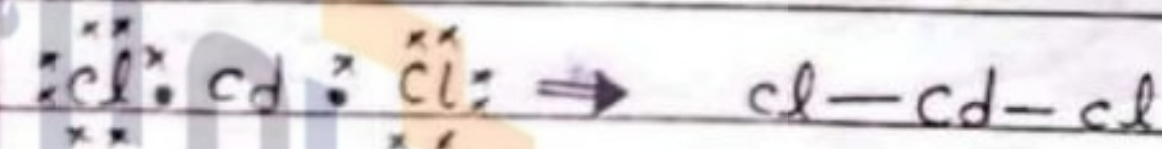
E.C of $Cl_{(17)}$:- $[Ne]_{10}, 3s^2, 3p_x^2, 3p_y^2, 3p_z^1$



(ii) $CdCl_2$:-

E.C of $Cd_{(48)}$:- $[Kr]_{36}, 4d^{10}, 5s^2$

E.C of $Cl_{(17)}$:- $1s^2, 2s^2, 2p^6, 3s^2, 3p_x^2, 3p_y^2, 3p_z^1$



(b) Molecules containing two electron pair with one lone pair:-

Symbol:- ABE

No. of electron pair = 2

No. of bond pair:- 1

No. of lone pair:- 1

Lewis Structure:- $\cdot A \cdot \times B$

VSEPR Structure:- $\cdot A - B$

Bond angle:- 180°

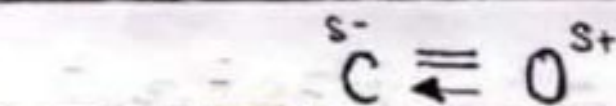
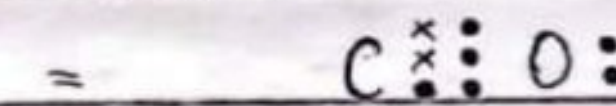
Geometry:- Linear

Example:-

CO :-

E.C of $C_{(6)}$:- $1s^2, 2s^2, 2p_x^2, 2p_y^1, 2p_z^1$

E.C of $O_{(8)}$:- $1s^2, 2s^2, 2p_x^2, 2p_y^1, 2p_z^1$



Type II:-

(a) Molecules containing three electron pair with no lone pair:-

Symbol:- AB_3

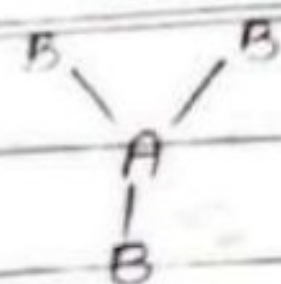
No. of electron pair:- 3

No. of bond pair:- 3

No. of lone pair:- 0

Lewis Structure:- $\begin{array}{c} B \cdot \times \cdot B \\ \times \cdot A \cdot \times \\ \times \end{array}$

VSEPR Structures:-



Bond angle:- 120°

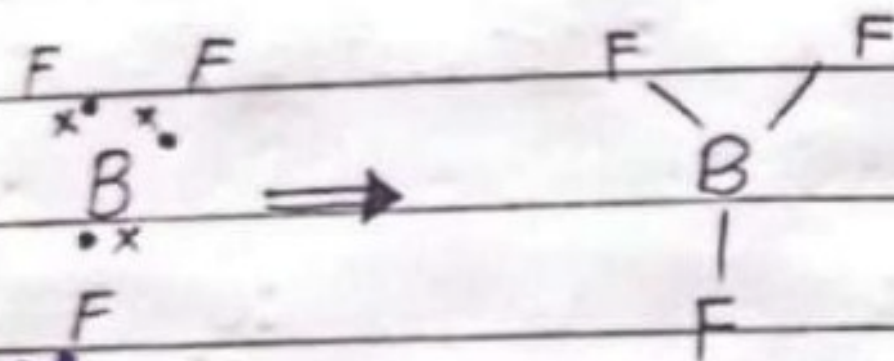
Geometry:- Triangular Planar.

Example:- BF_3 , AlCl_3

(i) BF_3 :-

F.C of B:- $1s^2, 2s^2, 2p_x^1, 2p_y^1, 2p_z^1$

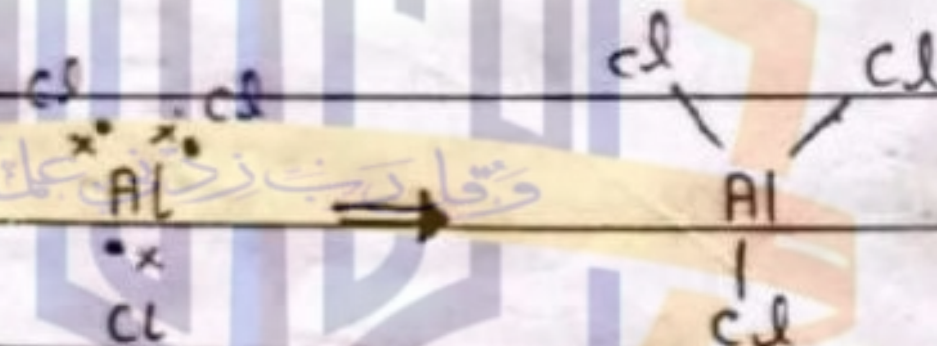
E.C of F:- $1s^2, 2s^2, 2p_x^2, 2p_y^2, 2p_z^1$
 $= [\text{He}], 2s^2, 2p_x^2, 2p_y^2, 2p_z^1$



(ii) AlCl_3 :-

F.C of Al:- $1s^2, 2s^2, 2p^6, 3s^2, 3p_x^1, 3p_y^1, 3p_z^1$

E.C of Cl:- $1s^2, 2s^2, 2p^6, 3s^2, 3p_x^2, 3p_y^2, 3p_z^1$



(b). Molecules containing three electron pair with one lone pair and Multiple bond:-

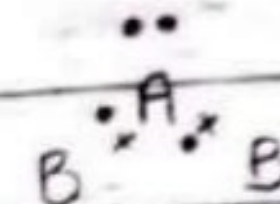
Symbol:- AB_2E

No. of electron pair:- 3

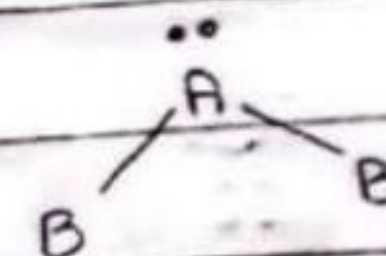
No. of bond pair:- 2

No. of lone pair:- 1

Lewis Structure:-



VSEPR Structure:-



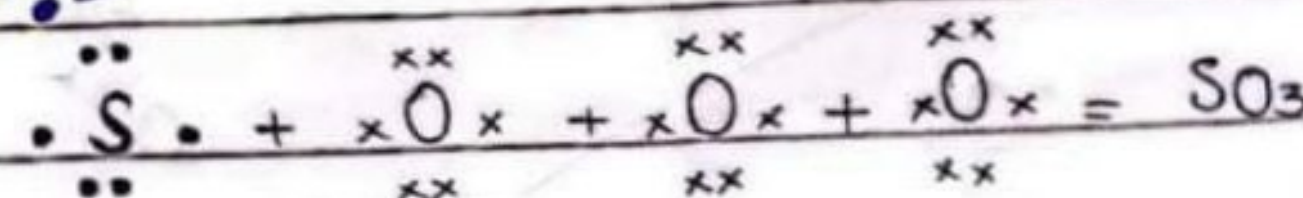
Bond angle:- 120°

Geometry:- angular or V-shaped

Examples (with lone pair and Multiple bond):-

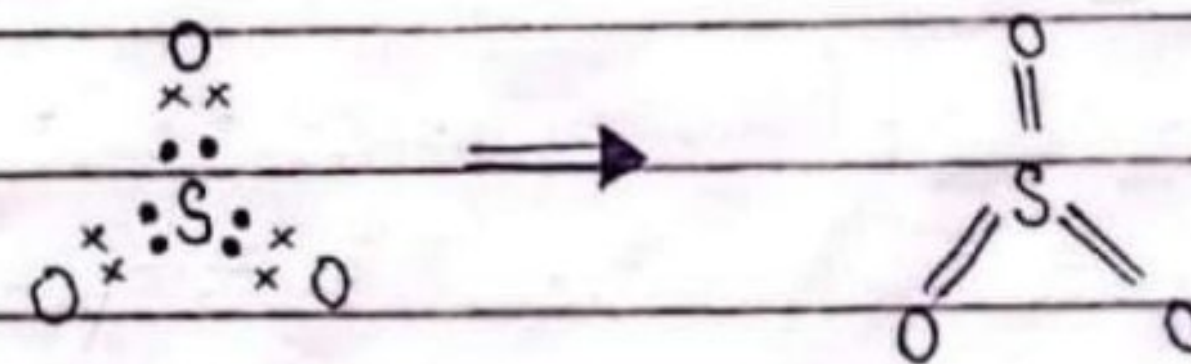
SO_3 , NO_3^- (Nitrate), CO_3^{2-} (carbonate),
 BO_3^{3-} (Borate)

(i) SO_3 :-



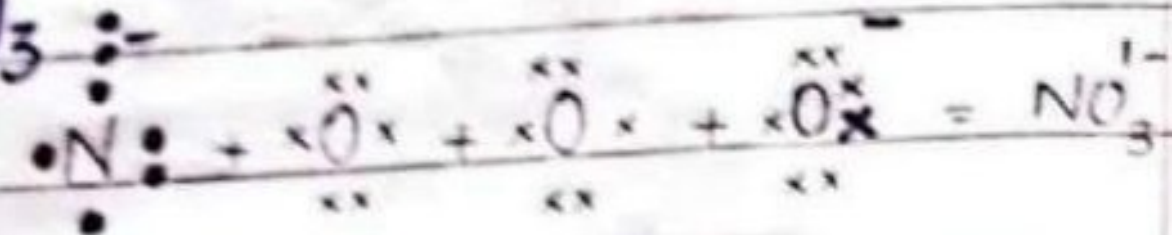
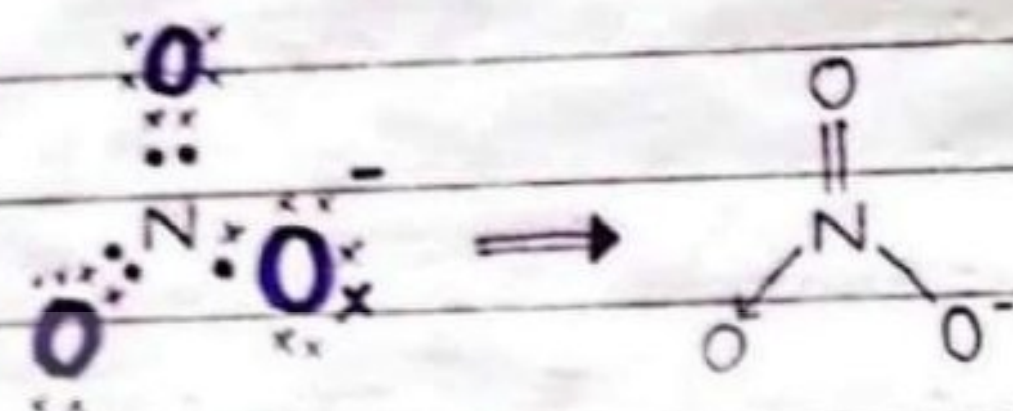
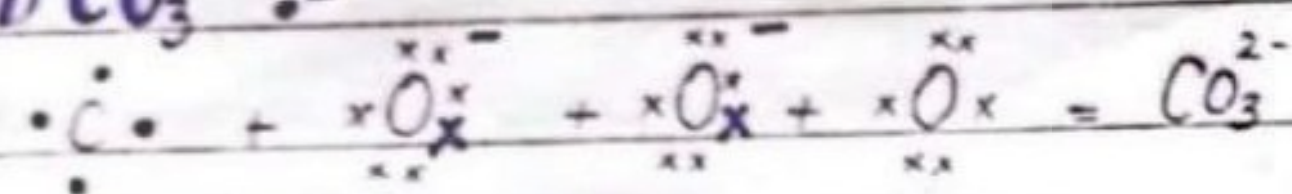
F.C of S:- $[\text{Ne}], 3s^2, 3p_x^2, 3p_y^1, 3p_z^1$

E.C of O:- $1s^2, 2s^2, 2p_x^2, 2p_y^1, 2p_z^1$



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(ii) NO_3^- :-E.C of N_{central} :- $1s^2, 2s^2, 2p_x^1, 2p_y^1, 2p_z^1$ E.C of O_{terminal} :- $1s^2, 2s^2, 2p_x^2, 2p_y^2, 2p_z^1$ (iii) CO_3^{2-} :-E.C of C_{central} :- $1s^2, 2s^2, 2p_x^1, 2p_y^1, 2p_z^1$ E.C of O_{terminal} :- $1s^2, 2s^2, 2p_x^2, 2p_y^2, 2p_z^1$ (iv) SnCl_2 :-E.C of Sn_{central} :- $[\text{Kr}]_{36}, 4d^{10}, 5s^2, 5p_x^1, 5p_y^1, 5p_z^0$ E.C of Cl_{terminal} :- $[\text{Ne}], 3s^2, 3p_x^2, 3p_y^2, 3p_z^1$ 

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Type III:-

(a) Molecules with four electron pair with no lone pair :-

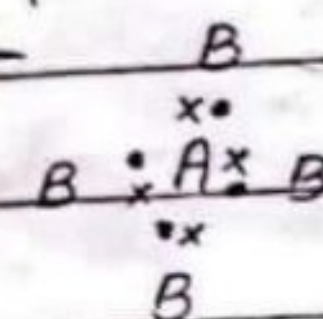
Symbol :- AB_4

No. of electron pair :- 4

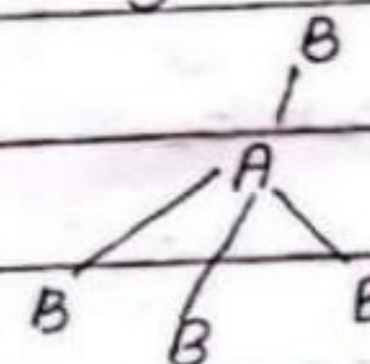
No. of bond pair :- 4

No. of lone pair :- 0

Lewis Structure :-



VSEPR Structure :-

Bond angle :- 109.5°

Geometry :- Tetrahedral.

A = Elements belonging to group

IVA i.e. :- C, Si, Ge.

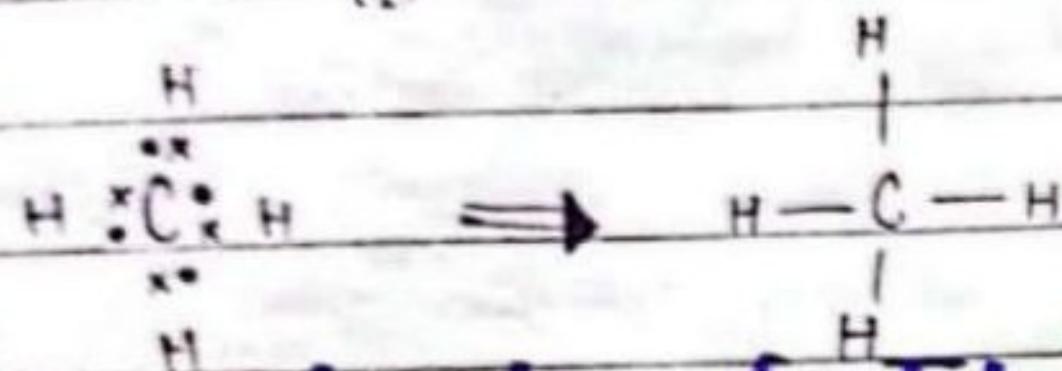
B = H, F, Cl, Br

Example :-

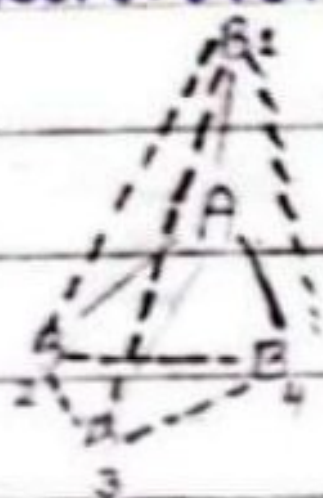
 $\text{CH}_4, \text{CCl}_4, \text{SiH}_4, \text{SiF}_4$ (i) CH_4 :-E.C of C_{central} :- $1s^2, 2s^2, 2p_x^1, 2p_y^1, 2p_z^1$

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E.C of $H_{(1)}$:- $1s^1$ 

Characteristic of Tetrahedral:-



Total Bond angle :- 6

Total edges :- 6

Total Corner :- 4

Total Faces :- 4

Bond angle:-

1. $m < B_2 A B_2$ 2. $m < B_1 A B_3$ 3. $m < B_1 A B_4$ 4. $m < B_2 A B_3$ 5. $m < B_2 A B_4$ 6. $m < B_3 A B_4$

Edges:-

= 6

Corner:-

= 4

Faces:-

(i) $B_1 B_2 B_3$ (ii) $B_1 B_2 B_4$ (iii) $B_1 B_3 B_4$ (iv) $B_2 B_3 B_4$

(b) Molecules containing four electron pair with one lone pair:-

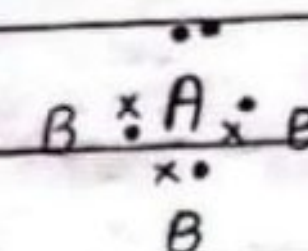
Symbol :- AB_3E

No. of electron pair :- 4

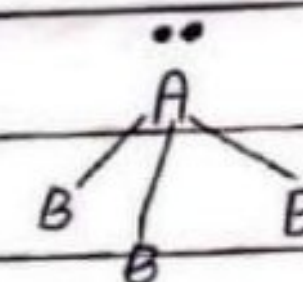
No. of bond pair :- 3

No. of lone pair :- 1

Lewis Structure:-

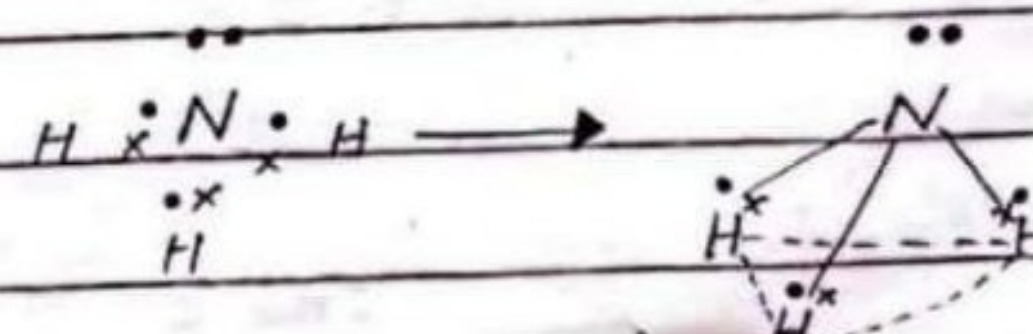


VSEPR Structure:-

Bond angle:- Less than 109.5°

Geometry:- Triangular pyramidal.

Example:-

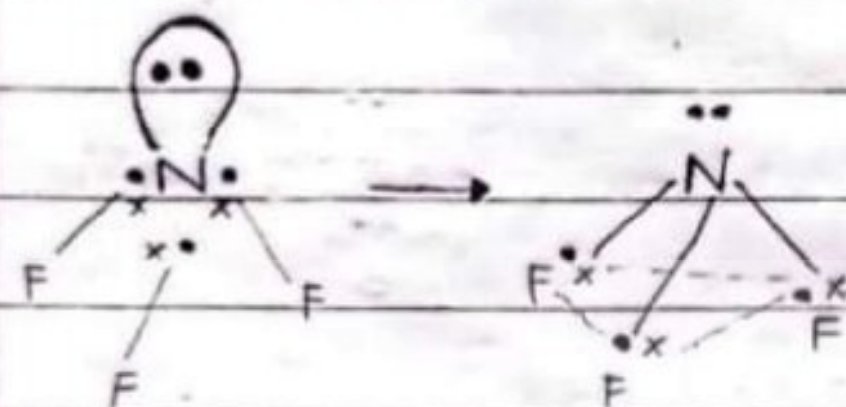
(i) NH_3 :-E.C of $N_{(7)}$:- $[He], 2s^2, 2p_x^1, 2p_y^1, 2p_z^1$ E.C of $H_{(1)}$:- $1s^1$ 

→ Effect of increasing electronegativity on $\angle B$ on bond angle.

(ii) NF_3 :-

→ F atom attracts bond pair towards their (repulsive) nucleus.

→ Lone pair is also drifted close to the nucleus of nitrogen.



Flourine is highly electronegative therefore electron pair is close to the Flourine.

(C). Molecules containing Four electron pairs, with two lone pairs:-

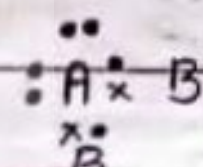
Symbol:- AB_2E_2

No. of electron pair:- 4

No. of bond pair:- 2

No. of lone pair:- 2

Lewis Structure:-

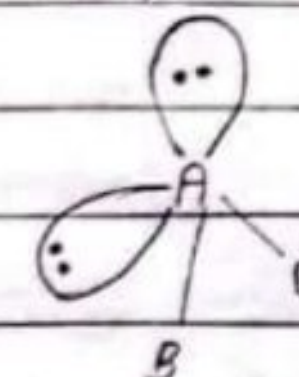


A = Elements belonging to group

IVA i.e.:- C, Si, Ge, etc.

B = H, Halogens (F, Cl)

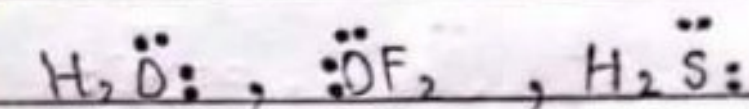
VSEPR Structure:-



Bond angle:- Far less than 109.5°

Shape:- V-shaped or angular or bent.

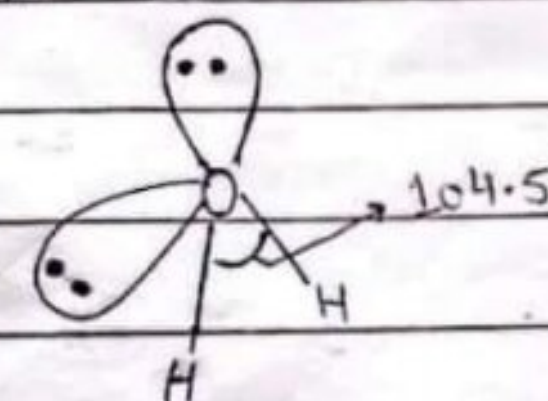
Example:-



(i) H_2O :-

E.C of O:- $[\text{He}], 2s^2, 2p_x^2, 2p_y^1, 2p_z^1$

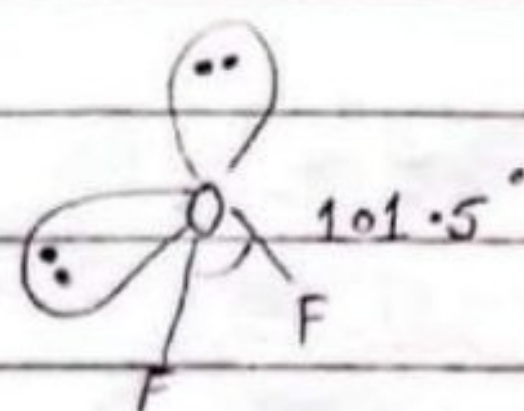
E.C of H:- $1s^1$



(ii) OF_2 :-

E.C of F:- $[\text{He}], 2s^2, 2p_x^2, 2p_y^2, 2p_z^1$

E.C of O:- $[\text{He}], 2s^2, 2p_x^2, 2p_y^1, 2p_z^1$



Type IV:-

(a). Molecules containing Five electron pair with no lone pair:-

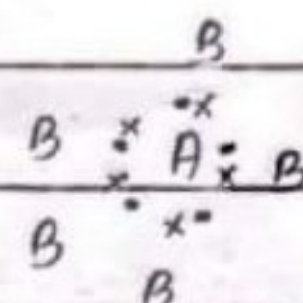
Symbol:- AB_5

No. of electron pair:- 5

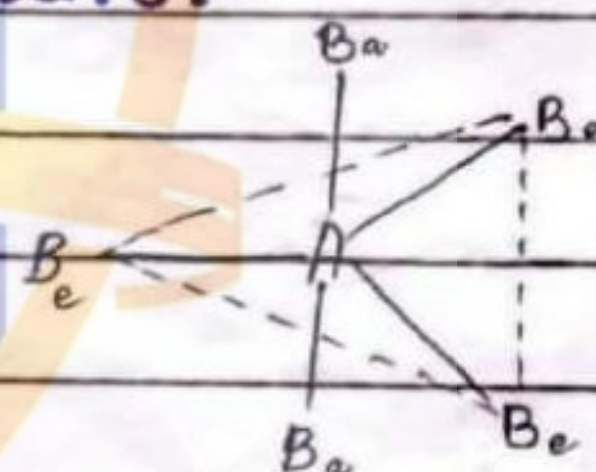
No. of bond pair:- 5

No. of lone pair:- 0

Lewis Structure:-



VSEPR Structure:-



Bond angle:-

(i) $m \angle B_e A B_e = 120^\circ$

(ii) $m \angle B_a A B_e = 90^\circ$

Geometry:- Trigonal Bipyramidal
Example:-

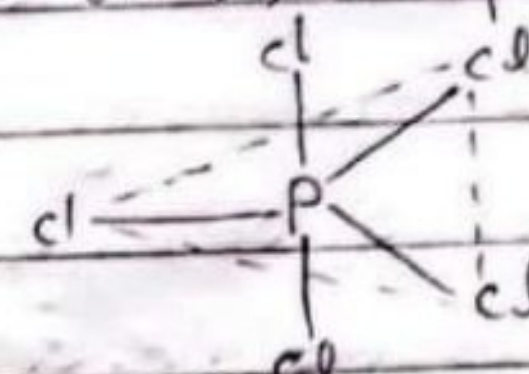
PCl_5 , PBr_5

A = Elements belonging to VA group i.e. N, P, As etc.
B = H, Halogens (F, Cl)

(i) PCl_5 :-

E.C of P(15) :- $[Ne], 3s^2, 3p_x^1, 3p_y^1, 3p_z^1$

E.C of Cl(17) :- $[Ne], 3s^2, 3p_x^2, 3p_y^2, 3p_z^1$



(b). Molecules containing Five electron pair with one lone pair:-

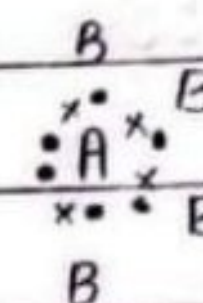
Symbol:- AB_4E

No. of electron pair:- 5

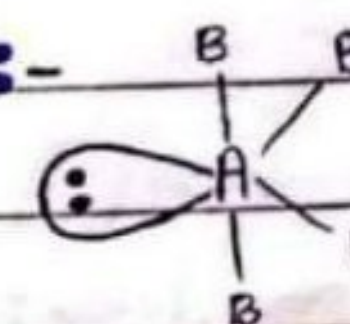
No. of bond pair:- 4

No. of lone pair:- 1

Lewis Structure:-



VSEPR Structure:-



Bond angle:-

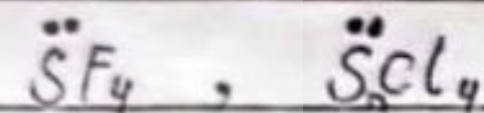
- (i) 120°
- (ii) Less than 90° .
- (iii) Less than 180° .

Geometry:- Sea saw or Distorted Tetrahedral.

A = Elements belonging to group

VIA i.e. S, Se, etc.

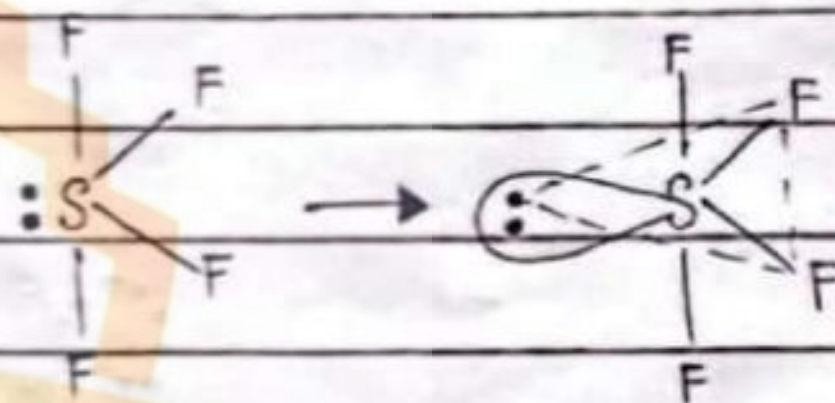
B = H, Halogens (F, Cl).

Example:-

(i) SF_4

E.C of S(16) :- $[\text{Ne}], 3s^2, 3p^2, 3p_y^1, 3p_z^1$

E.C of F(9) :- $[\text{He}], 2s^2, 2p_x^2, 2p_y^1, 2p_z^1$



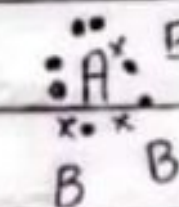
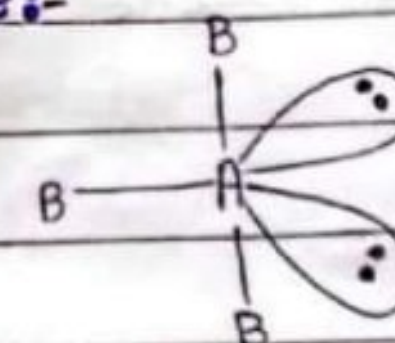
(c) Molecules with Five electron pair with two lone pair:-

Symbol:- AB_3E_2

No. of electron pair:- 5

No. of bond pair:- 3

No. of lone pair:- 2

Lewis Structure:-**VSEPR Structure:-****Bond angles:-**

- (i) Less than 90°
- (ii) Less than 180° .

Geometry:- T-shaped

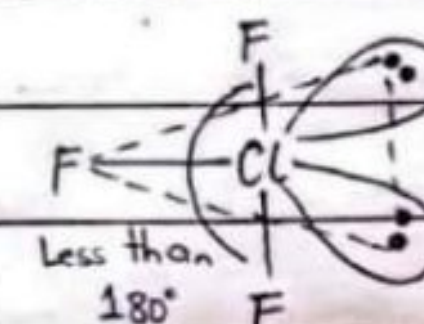
A = Elements belonging to group to VIIA. i.e. Cl, Br, I

F is not taken as central atom due to its smaller outer valence shell.

B = F, Br, Cl, ...

Example:- ClF_3

ClF_3 :-



(d). Molecules containing five electron pair with three lone pair:-

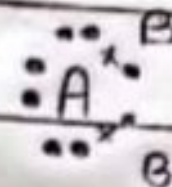
Symbol:- AB_2E_3

No. of electron pair:- 5

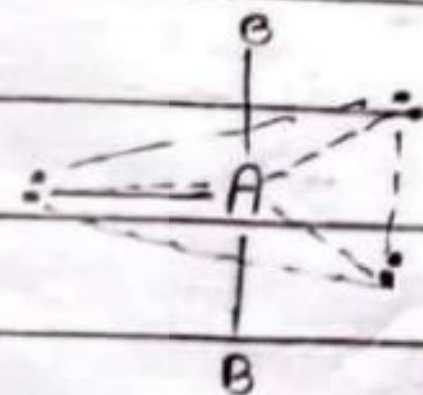
No. of bond pair:- 2

No. of lone pair:- 3

Lewis Structure:-



VSEPR Structure:-



Bond angle:- Less than 180° .

Geometry:- Linear.

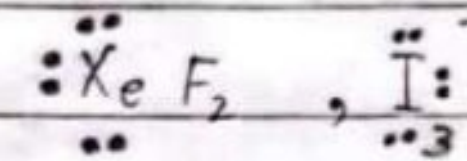
A = Elements belonging to group

VIIIA group (Noble gases), i.e.:- Kr, Xe, Rn.

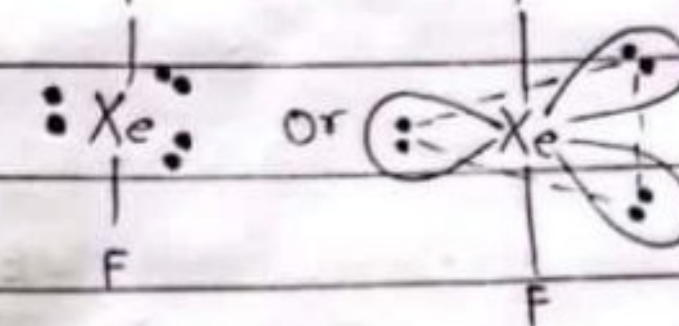
He, Ne, Ar, is not used taken as central atom due to its smaller outer valence shell.

B = F, Br, Cl

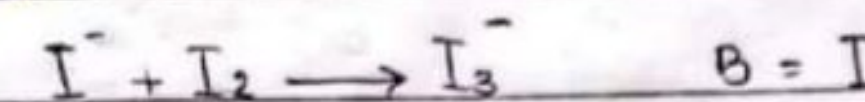
Example:-



(i) XeF_2 :- E.C of Xe:- $[Kr], 4d^{10}, 5s^2, 5p^6, 5d^0$

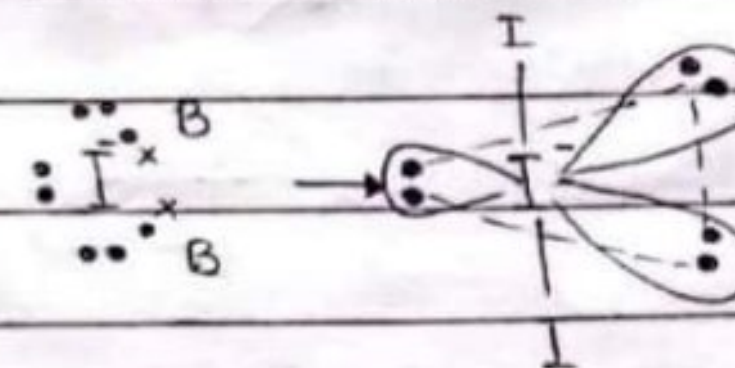


(ii) I_3^- :-



E.C of $I_{(53)}$:- $[Kr], 4d^{10}, 5s^2, 5p^6, 5d^0$

E.C of $I_{(54)}^-$:- $[Kr], 4d^{10}, 5s^2, 5p^6, 5d^0$



Type V:-

(a). Molecules containing six electron pair with no lone pair:-

Symbol:- AB_6

No. of electron pair = 6

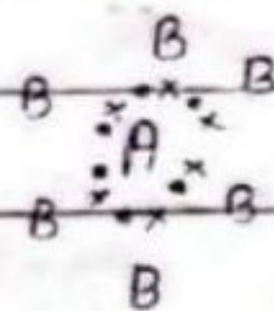
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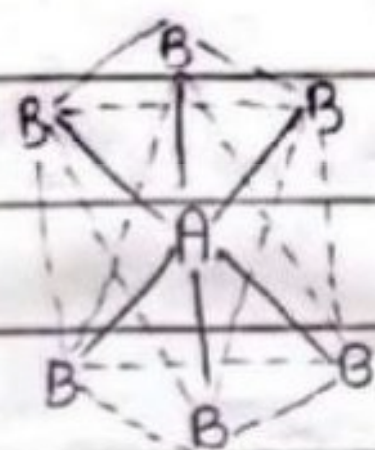
No. of bond pair: - 6

No. of lone pair: - 0

Lewis Structure:-



VSEPR Structure:-

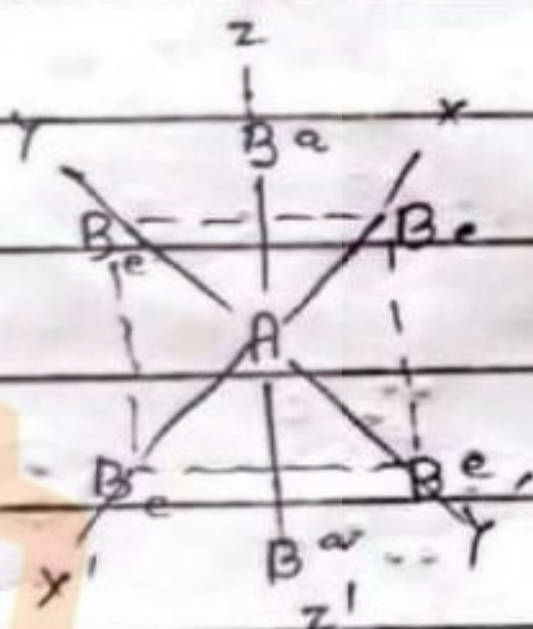


Bond angle:-

(i) $m \angle B_e A B_e = 90^\circ$

(ii) $m \angle B_a A B_e = 90^\circ$

(iii) $m \angle B_a A B_a = 180^\circ$



Geometry:- Octahedral (square Bipyramidal)

A = Elements belonging to group

VIA i.e. S, Se

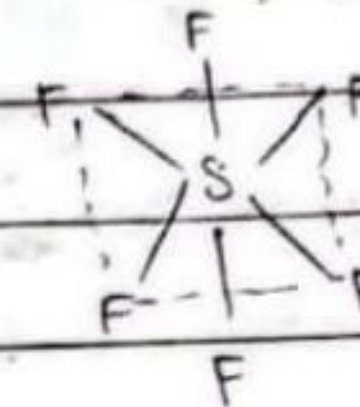
B = F, Cl, Br

Example:-

 SF_6 :-

DAY: _____

DATE: _____

E.C of $F(a)$:- $[He], 2s^2, 2p_x^2, 2p_y^2, 2p_z^1$ 

(b). Molecules containing Six electron with one lone pair:-

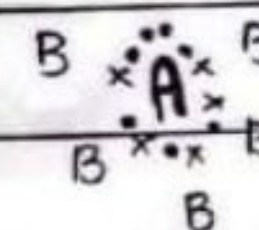
Symbol:- AB_5E

No. of electron pair:- 6

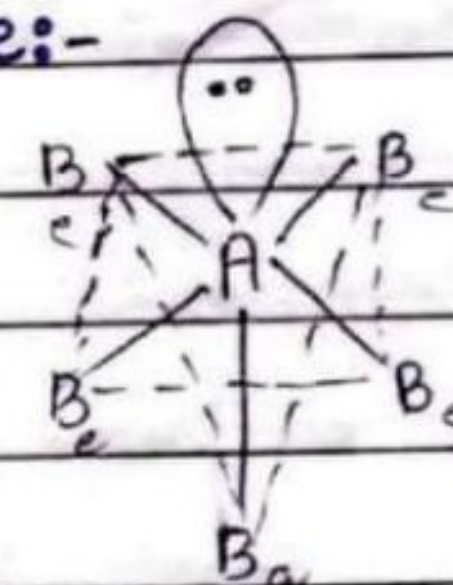
No. of bond pair:- 5

No. of lone pair:- 1

Lewis Structure:-



VSEPR Structure:-



Bond angle:-

(i) $m \angle B_e A B_e = 90^\circ$

(ii) $m \angle B_a A B_e = 90^\circ$

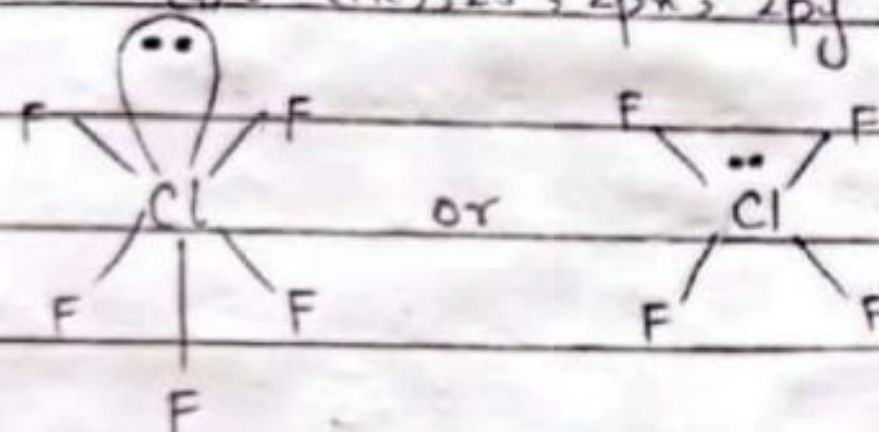
Geometry: Square pyramidal
 A = Elements belonging to group
 VIIA group i.e. Cl, Br, I, etc.
 B = F, Cl, Br, etc.

Example:-

ClF_5 , ICl_5

ClF_5 :-

E.C of Cl :- $(\text{Ne}), 3s^2, 3p^4, 3d^0, 3p^1$
 E.C of F :- $(\text{He}), 2s^2, 2p^4, 2d^0, 2p^1$



(C). Molecules containing Six
 electron pair with two
 lone pair:-

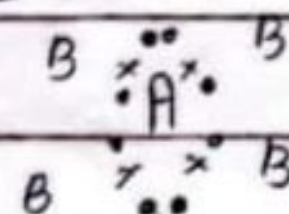
Symbol:- AB_4E_2

No. of electron pair:- 6

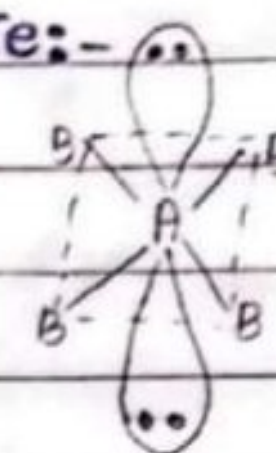
No. of bond pair:- 4

No. of lone pair:- 2

Lewis Structure:-



VSEPR Structure:-



Geometry:- Square planar

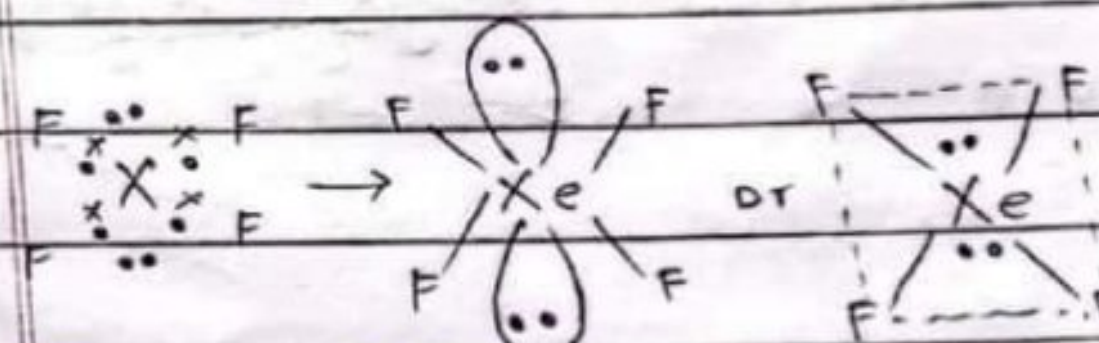
A = Elements must belonging to
 group VIIIA group i.e. Kr, Xe, etc.
 B = Cl, Br, F

Example:-

XeF_4 , KrCl_4 etc.

XeF_4 :-

$(\text{Xe})_{54} :- (\text{Kr})_{36}, 4d^{10}, 5s^2, 5p^6, 5d^0, 5p^2$



Type VI:-

(a). Molecules with Seven electron
 Pair with no lone Pair:-

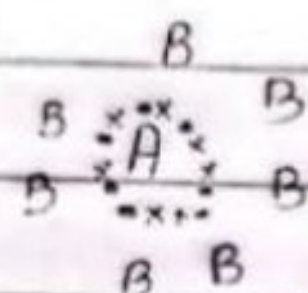
Symbol:- AB_7

No. of electron pair:- 7

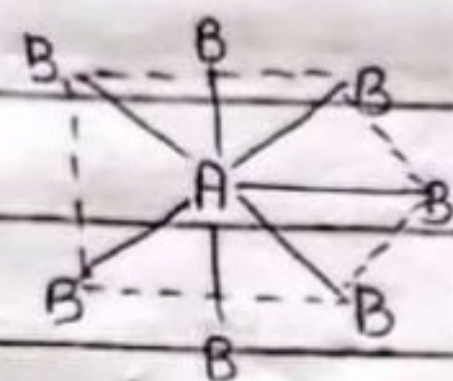
No. of bond pair:- 7

No. of lone pair: 0

Lewis Structure:-



VSEPR Structure:-



Bond angle:-

(i) $m\angle Be A Be = 108^\circ$

(ii) $m\angle Be A Be = 90^\circ$

(iii) $m\angle Ba A Ba = 180^\circ$

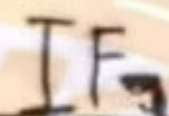
Geometry:- Pentagonal Bipyramidal.

A = Elements must belonging to group

VIIA i.e. Kr, Br, I, etc.

B = F, Cl, Br, etc.

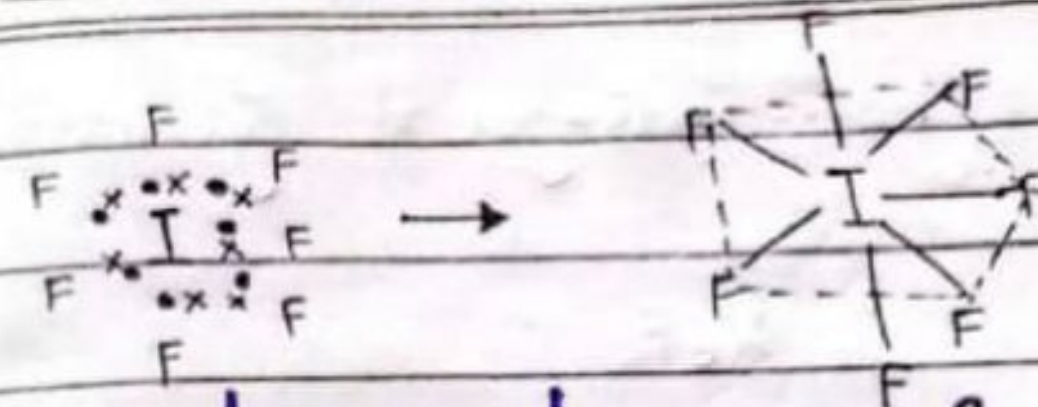
Example:-



IF_5 :-

E.C. of $I_{(53)}$:- $[Kr]_{36}, 4d^{10}, 5s^2, 5p^5, 5p_z^2, 5p_z^2$

E.C. of $F_{(9)}$:- $[He], 2s^2, 2p_x^2, 2p_y^2, 2p_z^1$



(b) Molecules containing Seven electron pair with One pair:-

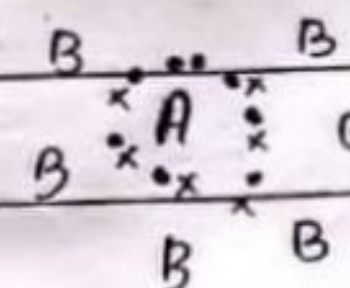
Symbol:- AB_6E

No. of electron pair = 7

No. of bond pair = 6

No. of lone pair = 1

Lewis Structure:-

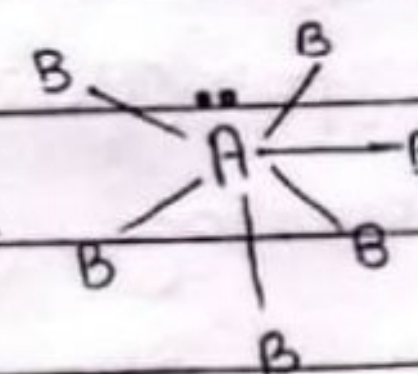


VSEPR Structure:-

A = Elements must

belonging to group VIIA.

i.e. Xe, Kr etc.



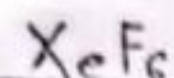
Bond angle:-

(i) $m\angle Be A Be = 108^\circ$

(ii) $m\angle Be A Be = 90^\circ$

(iii) $m\angle Ba A Ba = 180^\circ$

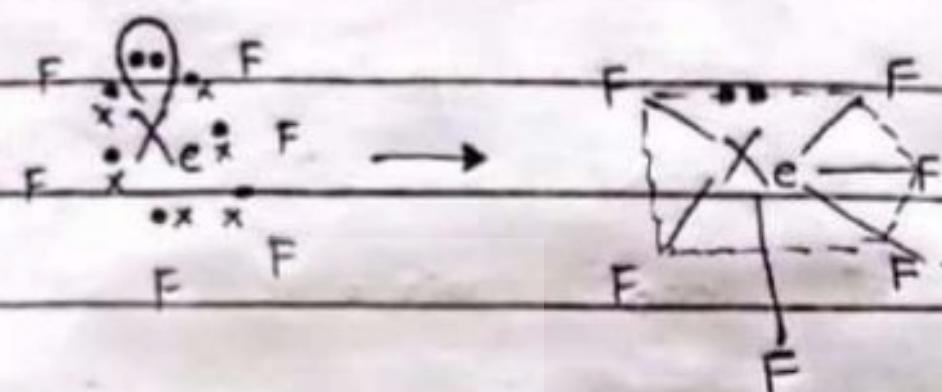
Geometry:- Pentagonal Pyramidal.
Example:-



XeF_6 :-

E.C of Xe :- $[\text{Kr}], 4d^{10}, 5s^2, 5p^6, 5d^0, 5p^6, 5p^2$

E.C of F :- $[\text{He}], 2s^2, 2p^6, 2p^2, 2p^2, 2p^2$



Limitation of VSEPR:-

1. VSEPR does not explain the shape of molecules that have inert pair (Electron not involved in bonding in certain cases).

e.g. In SnCl_4 (Stannic chloride)

In SnCl_2 (stannous chloride).

2. It does not explain the shape of transition elements.

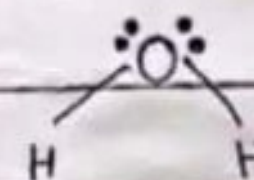
3. VSEPR does not explain the shape of molecules

that is d-localized.

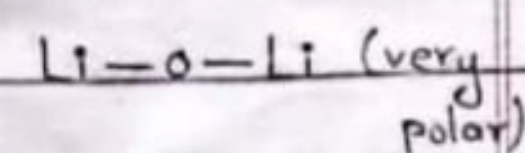
4. VSEPR does not explain the shape of molecules that have very polar bond.

e.g.:- In H_2O , Li_2O

$\Rightarrow \text{H}_2\text{O} :-$



$\Rightarrow \text{Li}_2\text{O} :-$



Shape:- angular

Shape:- Linear

E.N of H = 2.1

E.N of Li = 0.9

E.N of O = 3.5

E.N of O = 3.5

Difference = 1.4

Difference = 2.6

5. VSEPR does not explain the formation of covalent bond.